Solution to Quiz 2.

1. (a) Def: If $X: \Omega \to \mathbb{R}$ is $\mathcal{A}$-measurable, we call X is a random variable on $(\Omega, \mathcal{A}, \mathbb{P})$.

examples: $([0,1], \mathcal{B}[0,1], \mathbb{P}=$ Lebesgue measure$)$

$f(x) = 1_{[0,\frac{1}{2}]}$ $\quad$ $f(x) = x$

(b) Def: σ-algebra $\mathcal{A}_1$ and $\mathcal{A}_2$ are independent if $\forall B_1 \in \mathcal{A}_1$ and $B_2 \in \mathcal{A}_2$

$$P(B_1 \cap B_2) = P(B_1) \cdot P(B_2)$$

Def: Random Variables X_1 and X_2 are independent if

$\sigma(X_1)$ and $\sigma(X_2)$ are independent

(c) Def: Let $g: I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is a open interval. We say g is convex if

$$g(\alpha x + (1-\alpha) y) \leq \alpha g(x) + (1-\alpha) g(y), \quad \forall x, y \in I \quad \forall \alpha \in [0,1]$$

2. (a) ① F is non-decreasing. ② F is right continuous ③ $\lim_{x \to \infty} F(x) = 1$ and $\lim_{x \to -\infty} F(x) = 0$.

(b) T (c) T (d) T (e) T

3. ① X is a non-negative random variable, then

$$P(X \geq a) \leq E[X]/a.$$

proof: $P(X \geq a) = E[1_{\{X \geq a\}}] \leq E[X]/a$

② X is also a nonnegative random variable, then

$$P(|X| \geq a) \leq \frac{E[|X|^p]}{a^p}$$

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proof: By the Markov inequality,

$$P(|X|\geq a) = P(|X|^p \geq a^p) \leq E[|X|^p]/a^p$$

[4] $$\sum_{n=1}^{\infty} P(A_n) = \sum_{n=1}^{\infty} n^{-\alpha} = \begin{cases} +\infty & \alpha \leq 1 \\ <\infty & \alpha > 1 \end{cases}$$

By Lemma 3.1.8, we have $P(\limsup_{n\to\infty} A_n) = \begin{cases} 0 & \text{if } \alpha > 1 \\ 1 & \text{if } \alpha \leq 1 \end{cases}$

[5] (a) $$P(\min(X,Y) > i) = P(\{X > i\} \cap \{Y > i\})$$

$$= P(X > i)\, P(Y > i)$$

$$= \left(\sum_{k=i}^{\infty} \frac{1}{2^{k+1}}\right)^2 = \left(\frac{1}{2^i}\right)^2$$

Thus $P(\min(X,Y) \leq i) = 1 - \frac{1}{4^i}$

(b) $$P(X = Y) = P\left(\bigcup_{i=1}^{\infty} \{X = Y = i\}\right)$$

$$= \sum_{i=1}^{\infty} P(X = i, Y = i)$$

$$= \sum_{i=1}^{\infty} P(X = i)\, P(Y = i)$$

$$= \sum_{i=1}^{\infty} \frac{1}{4^i} = \frac{1}{3}$$

(c) By symmetry,

$$P(X > Y) = P(X < Y)$$

So. $P(X > Y) + P(X < Y) + P(X = Y) = 1.$

Hence, $P(X < Y) = \frac{1 - \frac{1}{3}}{2} = \frac{1}{3}.$

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